Slick Trace investigation report — case_0003

Generated 2026-09-26T13:00:36.354496+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

Leading candidate: 419771272. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

Scene

Field	Value
Incident	case_0003
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
candidate	5

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.925 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	13dd40672ab6431e
coverage_result.json	6ceb13a8c2e6da7d
forcing.npz	767ec8f532e8cc46
forecast_result.json	efb953534425a760
fragility_result.json	42a592bcdf7fe525
historical_mapping.json	748c15129fabdb2c
manifest.json	1ff45cd57416e0cd
mask_groundtruth.png	f68e46a604d4c3c2
naive_baseline_result.json	ec0c618119c3a020
next_observation_result.json	ae115bc6f214ab9b
scene.png	2e9edc714c5336e2

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.